# A DECENT ALGEBRAIC SPACE HAS A DENSE OPEN SUBSPACE WHICH IS A SCHEME

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*Editorial note.* No omission markers in-span.

*Allowed standard prerequisites (editorial).*

- Largest open subscheme. Properties of Spaces, lemma subscheme (cross-chapter, named). Sets up the contradiction reduction.
- Degree filtration. lemma-stratify-flat-fp-lqf (carded L26711). The open sequence X = X_0 superset X_1 superset ...
- Reasonable implies dense open scheme. proposition-reasonable-open-dense-scheme (carded L26717). The contradiction in the inductive choice of V_n.
- Nonempty limits of affine opens. Limits, lemma limit-nonempty (cross-chapter, named). The limit V_infinity is nonempty.

*Source reference.* Original lemma 086T: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1715–1715. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 01Z2: focused-seeds/geometry/S9-decent-decent-open-dense-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original proposition 03JI: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2273–2273. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03JH: focused-seeds/geometry/S9-decent-decent-open-dense-scheme/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Theorem 0.1** (David Rydh)**.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. If $X$ is decent, then there exists a dense open subspace of $X$ which is a scheme.*

*Proof.* Assume $X$ is a decent algebraic space for which the theorem is false. By Properties of Spaces, Lemma 03JH there exists a largest open subspace $X' \subset X$

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which is a scheme. Since $X'$ is not dense in $X$, there exists an open subspace $X'' \subset X$ such that $|X''| \cap |X'| = \emptyset$. Replacing $X$ by $X''$ we get a nonempty decent algebraic space $X$ which does not contain *any* open subspace which is a scheme.

Choose a nonempty affine scheme $U$ and an étale morphism $U \to X$. We may and do replace $X$ by the open subscheme corresponding to the image of $|U| \to |X|$. Consider the sequence of open subspaces

$$X = X_0 \supset X_1 \supset X_2 \ldots$$

constructed in Lemma 086T for the morphism $U \to X$. Note that $X_0 = X_1$ as $U \to X$ is surjective. Let $U = U_0 = U_1 \supset U_2 \ldots$ be the induced sequence of open subschemes of $U$.

Choose a nonempty open affine $V_1 \subset U_1$ (for example $V_1 = U_1$). By induction we will construct a sequence of nonempty affine opens $V_1 \supset V_2 \supset \ldots$ with $V_n \subset U_n$. Namely, having constructed $V_1, \ldots, V_{n-1}$ we can always choose $V_n$ unless $V_{n-1} \cap U_n = \emptyset$. But if $V_{n-1} \cap U_n = \emptyset$, then the open subspace $X' \subset X$ with $|X'| = \mathrm{Im}(|V_{n-1}| \to |X|)$ is contained in $|X| \setminus |X_n|$. Hence $V_{n-1} \to X'$ is an étale morphism whose fibres have degree bounded by $n - 1$. In other words, $X'$ is reasonable (by definition), hence $X'$ contains a nonempty open subscheme by Proposition 03JI. This is a contradiction which shows that we can pick $V_n$.

By Limits, Lemma 01Z2 the limit $V_\infty = \lim V_n$ is a nonempty scheme. Pick a morphism $\mathrm{Spec}(k) \to V_\infty$. The composition $\mathrm{Spec}(k) \to V_\infty \to U \to X$ has image contained in all $X_d$ by construction. In other words, the fibred $U \times_X \mathrm{Spec}(k)$ has infinite degree which contradicts the definition of a decent space. This contradiction finishes the proof of the theorem. $\square$